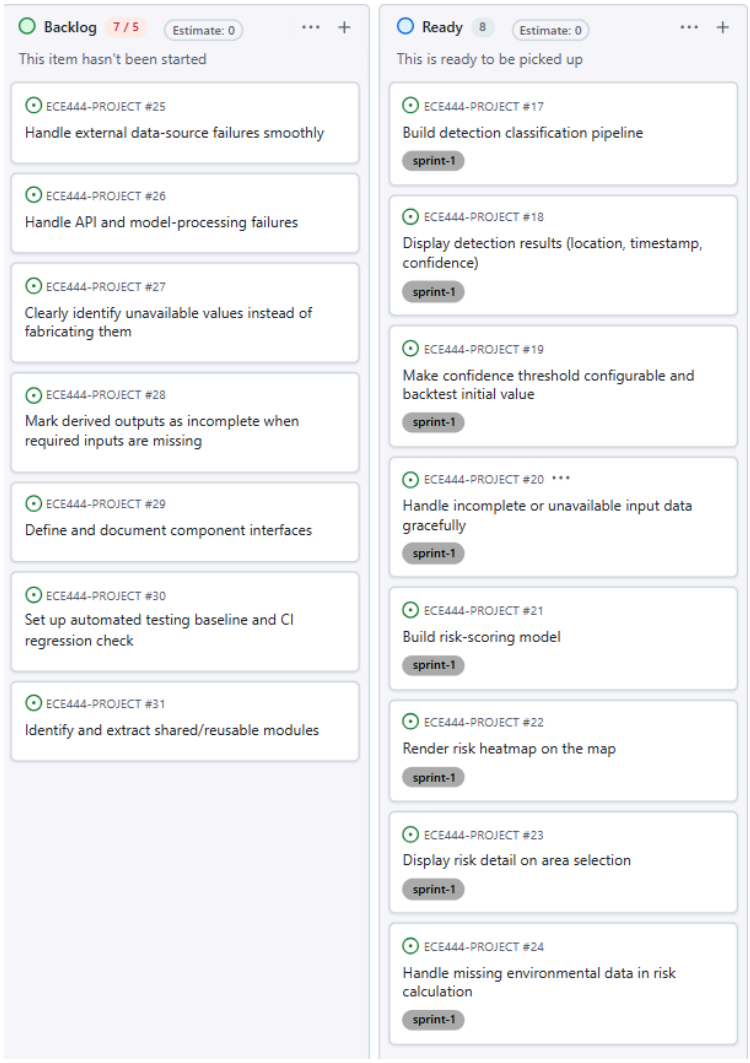
Team Ash Detectives – Milestone 3.1: First Sprint Task Planning

Team Ash Detectives (wildfire early-detection and risk-monitoring platform) tracks work in GitHub Projects; Sprint 1 covers detection and risk heatmap, Sprint 2 covers reliability and maintainability.

Project management tool and sorted backlog

Tool: **GitHub Projects** (Kanban board linked to our repository), with each implementation issue tracing back to a user story issue.

The backlog is sorted by priority. Detection (#10) and Heatmap (#3) sit in **Ready** for Sprint 1 because risk scoring and detection feed every other feature. Reliability (#4) and Maintainability (#12) stay in **Backlog** until there is working functionality to harden.



Sprint 1 summary

Sprint 1 focused on the two stories every other feature depends on: **Wildfire Detection and Confidence Scoring (#10)** and **Wildfire Risk Heatmap (#3)**. Each story was split into 4 implementation issues (8 total), all moved to Ready on the GitHub Projects board.

- **Detection (#10):** classification pipeline using NASA FIRMS data, display of location/timestamp/confidence, configurable confidence threshold with an initial backtest, and graceful handling of missing input data.
- **Heatmap (#3):** risk-scoring model (#21), heatmap rendering on the map (#22), risk detail on area selection (#23), and handling of missing environmental data (#24).
- **Deferred to Backlog:** Reliability (#4, 4 issues) and Maintainability (#12, 3 issues), since they need working features to apply to.
- **Process:** pull requests are reviewed by a teammate before merging.

Sprint 2 plan

Sprint 2 pulls the 7 backlog issues into scope and finishes anything that carried over from Sprint 1.

- **Reliability (#4, 4 issues):** external data-source failures (#25), API and model-processing failures (#26), clearly identified unavailable values (#27), and outputs marked incomplete when inputs are missing (#28).
- **Maintainability (#12, 3 issues):** defined and documented component interfaces (#29), automated testing baseline with a CI regression check (#30), and extraction of shared modules (#31).
- **Carry-over:** any unfinished Sprint 1 detection or heatmap issues.
- **Ongoing:** timely pull request reviews, team meetings, and test cases for the Sprint 1 stories.

Tasks

Mahrous's tasks for the 1st sprint

Task	Estimated Time (hour)	Actual Time (hour)
Build detection classification pipeline (#17)	6	8

Display detection results: location, timestamp, confidence (#18)	3	5
Add unit tests for #17 and #18	2	2
Review team-member pull requests timely and thoroughly	2	2
Participate actively in common team duties (e.g., meetings)	2	2

Mahrous's tasks for the 2nd sprint

Task	Estimated Time (hour)	Actual Time (hour)
Handle external data-source failures smoothly (#25)	3	
Handle API and model-processing failures (#26)	3	
Finish any carry-over Sprint 1 issues	2	
Add test cases for #25 and #26	2	
Review team-member pull requests timely and thoroughly	2	

Participate actively in common team duties (e.g., meetings)	2	

Malhotra's tasks for the 1st sprint

Task	Estimated Time (hour)	Actual Time (hour)
Render risk heatmap on the map (#22)	5	7
Handle incomplete or unavailable input data gracefully (#20)	2	4
Set up repository structure and coding standards as Software Lead	2	2
Review team-member pull requests timely and thoroughly	2	2
Participate actively in common team duties (e.g., meetings)	2	2
Write README file	2	1

Malhotra's tasks for the 2nd sprint

Task	Estimated Time (hour)	Actual Time (hour)
Define and document component interfaces (#29)	3	
Set up automated testing baseline and CI regression check (#30)	4	
Add test cases for #22 and #20	2	
Review team-member pull requests timely and thoroughly	3	
Participate actively in common team duties (e.g., meetings)	2	

Cheralody's tasks for the 1st sprint

Task	Estimated Time (hour)	Actual Time (hour)
Display risk detail on area selection (#23)	4	5
Handle missing environmental data in risk calculation (#24)	3	3

Sort backlog and maintain the GitHub Projects board	2	2
Write Contribution file	2	2
Review team-member pull requests timely and thoroughly	2	2
Participate actively in common team duties (e.g., meetings)	3	2

Cheralody's tasks for the 2nd sprint

Task	Estimated Time (hour)	Actual Time (hour)
Clearly identify unavailable values instead of fabricating them (#27)	2	
Identify and extract shared/reusable modules (#31)	3	
Update the board and write the sprint report	3	
Add test cases for #23 and #24	2	
Review team-member pull requests timely and thoroughly	2	

Participate actively in common team duties (e.g., meetings)	3	

Samanta Akbar's tasks for the 1st sprint

Task	Estimated Time (hour)	Actual Time (hour)
Make confidence threshold configurable and backtest initial value (#19)	4	6
Build risk-scoring model (#21)	5	6
Prepare and clean NASA FIRMS data for model input	3	5
Write Code of Conduct file	2	1
Review team-member pull requests timely and thoroughly	2	2
Participate actively in common team duties (e.g., meetings)	2	2

Samanta Akbar's tasks for the 2nd sprint

Task	Estimated Time (hour)	Actual Time (hour)
Mark derived outputs as incomplete when required inputs are missing (#28)	2	
Improve model using Sprint 1 evaluation results	4	
Add test cases for #19 and #21 (model integration)	3	
Review team-member pull requests timely and thoroughly	2	
Participate actively in common team duties (e.g., meetings)	2	